Pranav Sunil Raja

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PROFILE

Data Science & AI professional delivering actionable insights and AI impact from complex data. Highly adaptable, results-oriented, and driven to solve real-world problems in fast-paced, growth-focused teams.

EDUCATION

MSc. Data Science (with Specialization in AI) **Dec 2025**
Newcastle University. *Newcastle Upon Tyne, United Kingdom.*
B.E Electronics and Communication **May 2024**
Visvesvaraya Technological University. *Karnataka, India.*

WORK EXPERIENCE

Dissertation Project, Open-Source OCR for Financial Document Processing **Feb 2025 – Aug 2025**
SAGE, United Kingdom.

- Conducted large-scale benchmarking of EasyOCR, Tesseract, and TrOCR as open-source alternatives to Azure OCR on financial and handwritten document datasets.
- Developed an end-to-end OCR and evaluation pipeline using PyTorch and Hugging Face, incorporating advanced image preprocessing and layout aware feature extraction.
- Applied NLP-based semantic post-processing using SBERT embeddings and fuzzy string matching to improve entity-level accuracy in financial data extraction.
- Designed a confidence-aware hybrid inference framework optimized using CER, WER, inference latency, and GPU memory profiling to balance model performance and deployment cost.

PROJECTS

Automated Claims Processing Using Databricks Lakehouse.

- Designed an end-to-end Medallion architecture on Databricks, ingesting batch and streaming data from SQL Server (CDC via Lakeflow Connect), Amazon S3, and AWS Kinesis using Lakeflow declarative pipelines with automated data quality checks and Unity Catalog-based governance.
- Implemented a computer vision pipeline by fine-tuning a ResNet model for car-damage severity classification, managing experiments with MLflow and Mosaic AI, and deploying real-time model serving integrated with business rules to automate insurance claim decisions.
- Delivered a full-stack analytics application using Databricks Apps (Streamlit) and Lakebase for low-latency OLTP access, enabling customer claim submission, internal investigations, and executive reporting through AI/BI dashboards and natural-language querying.

Scalable Healthcare Data Engineering Platform for Analytics and Benchmarking

- Designed and implemented a multi-tenant healthcare data ingestion pipeline using Apache Airflow, Snowflake, and Docker to standardize heterogeneous CMS datasets into a unified analytical schema.
- Built dynamic SQL-based data mapping and orchestration logic to process wide-format and line-item medical records through a single reusable DAG, enabling rapid onboarding of new datasets without custom pipelines.
- Integrated automated data quality validation checks (date-range auditing, record ID integrity, duplication detection) to ensure reliability of claims, beneficiary, and provider data for downstream analytics and benchmarking.

SKILLS

Programming and Libraries: Python, PySpark, SQL (MySQL), C++, R Programming, C Programming, NumPy, Pandas, Pytorch, Scikit-learn, OCR, NLP, Matplotlib, Seaborne.
Databases & ETL: MySQL, Data Warehousing, ETL Pipeline Development, CI/CD.
Big Data & Cloud: Fundamentals of AWS, Snowflake, Databricks, Hadoop, Google Colab.
Tools Proficiency: Power BI, Jupyter Notebook, Git, MATLAB, Microsoft Office.
Other: Data Pipeline Optimization, Report Automation (Power BI), Production Support & Troubleshooting.